ELEC70075
EE9-FPN2-11

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

EEE PART IV: MEng and ACGI

POWER SYSTEM ECONOMICS

Monday, 28 April 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Question 1

(a) Describe the significance of the concept of system marginal price and explain why it varies with time in the case of electricity.

[2]

(b) Three generators are available to supply a demand of $D = 600\ [MW]$. The cost of generating power $C_i(P_i)$ corresponding to each of the generators i is:

$$C_1(P_1) = 800 + 2 \cdot P_1 + 0.05 \cdot P_1^2\ [\frac{£}{h}]$$
$$C_2(P_2) = 650 + 8 \cdot P_2 + 0.25 \cdot P_2^2\ [\frac{£}{h}]$$
$$C_3(P_3) = 500 + 16 \cdot P_3 + 0.55 \cdot P_3^2\ [\frac{£}{h}]$$

Calculate the optimal production of each generator, the system marginal cost, the system average cost, and the profits of each generator.

[6]

(c) How would the dispatch change if three generators have the opportunity to buy some of this energy on the spot market at a price of 42.60 £/MWh? How much energy will be bought from the spot market?

[3]

(d) Assume that the minimum and maximum output limit of each generator is:

$$P_1^{min} = 100\ [MW],\ P_1^{max} = 600\ [MW]$$
$$P_2^{min} = 60\ [MW],\ P_2^{max} = 120\ [MW]$$
$$P_3^{min} = 40\ [MW],\ P_3^{max} = 100\ [MW]$$

Determine the optimal production of each generator, the system marginal cost, the system average cost and the profits of each generator. The spot market energy availability is neglected for calculation. Based on the results, briefly explain the impact of considering these maximum output limits on the total cost of operating the system and the profits of the three generators.

[6]

(e) In the solution of **(d)**, the profit of one of the generators may become negative. Explain why this happens, what this means for this generator, as well as why and how it should be avoided in practice.

[3]

Question 2

(a) List and explain key objectives of unbundling and liberalisation of electricity markets. Explain why producers taking part in a pool market do not have any incentives to bid at a price different to their marginal cost of production.

[4]

(b) Consider a market for electrical energy that is supplied by two generating companies with the two corresponding cost functions:

$$C_A(P_A) = 35 \cdot P_A + 0.37 \cdot P_A^2 \ [\frac{£}{h}]$$

$$C_B(P_B) = 39 \cdot P_B + 0.25 \cdot P_B^2 \ [\frac{£}{h}]$$

The inverse demand curve for this market is estimated to be:

$$\pi = 125 - 1.7 \cdot D \ [\frac{£}{MWh}]$$

where π and D are the electricity price and demand respectively. Assuming perfect competition, calculate the electricity price, level of demand, production levels and the profits made by the generating companies.

[4]

(c) Assuming a Cournot model of competition:
(i) Form a table to calculate the Nash equilibrium (NE) point of this market, i.e. market price, demand quantity and profit of each company for different levels of productions for each of the companies, as presented in Table 2.1

Table 2.1 Key parameters for assessing Nash Equilibrium

D (Demand)	Ω_A (profit A)
Ω_B (profit B)	π (Price)

Consider range of productions of company A at 14MW, 15MW and 16MW and of company B at 15MW, 16MW and 17MW.

[10]

(ii) Based on the results, briefly explain the impacts of imperfect competition on the market outcome.

[2]

Question 3

For the system shown in Figure 1, and the accompanied data given in Tables 1, 2 and 3, answer the following questions:

(a) Ignoring the impact of transmission constraints, calculate the minimum cost dispatch of the generators, the corresponding network flows and show that branch 1-2 would be overloaded.

[5]

(b) Change the dispatch by increasing generation at 3 to eliminate this overload and calculate the marginal price at 2.

[5]

(c) In order to manage the exposure to price fluctuations Load at 2 has (i) made a contract for difference with generation at 1 at a strike price of 26.5£/MWh and (ii) purchased a transmission congestion contract that gives the right to transport 90MW from bus 1 to bus 2. Calculate the income from the transmission congestion contract received by the Load at 2 and calculate the unit price that demand will pay for the energy consumed under this condition.

[5]

(d) Because of the maintenance on one of the lines of the transmission circuit between nodes 2 and 3, the flow through this circuit has to be restricted to 67MW. Calculate the new price at node 2. How much the demand at bus 2 will now pay for the electricity consumed, given the new conditions and its contract portfolio?

[5]

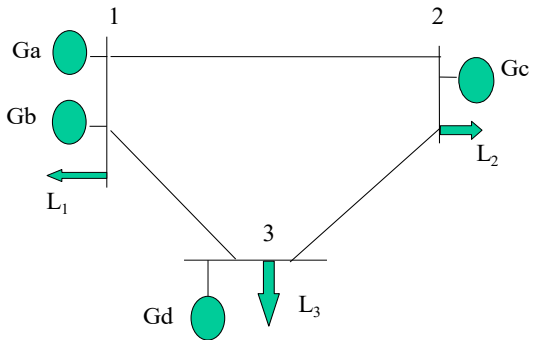


Figure 1: The system layout

Table 1 Line data

Line	Per unit reactance	Capacity (MW)
1-2	0.3	160
1-3	0.3	260
2-3	0.2	110

Table 2 Generator data

Generator	Capacity (MW)	Marginal cost (£/MWh)
Ga	170	19
Gb	345	15
Gc	110	35
Gd	105	25

Table 3 Load data

Node	Load (MW)
1	80
2	90
3	330

Question 4

(a) Two areas, A and B, of the Borduria power system are connected by a 1200MW transmission link operated by the Borduria Transco. System load is concentrated in area B. In winter, the load is 3,500MW while summer load is 2,000 MW. The cost of generation in the areas can be modelled by the following expressions:

$C(P_A) = 1 \cdot P_A + 0.001 \cdot P_A^2$ [£/h] for area A
$C(P_B) = 2 \cdot P_B + 0.002 \cdot P_B^2$ [£/h] for area B

Determine the optimal levels of generation in areas A and B for each of the seasons neglecting the impact of the transmission system.

[4]

(b) If necessary, modify the levels of generation computed in (a) to take into consideration the capacity of the transmission link. What are the electricity prices in areas A and B in each of the seasons? Calculate the generation cost in areas A and B for each of the seasons.

[4]

(c) Assuming that the duration of the winter and summer period is 2,700h and 6,060h respectively, calculate:

(i) The total annual generation costs

[2]

(ii) The total annual revenue received by the generators and the charge imposed on demand if electricity is priced at locational short-run marginal prices.

[2]

(iii) The total annual transmission revenue received by Borduria Transco in this case.

[2]

(d) Borduria Transco has proposed to the Borduria Regulator to reinforce the line between the two areas by 900 MW. Assuming that the annuitised investment cost of the additional line is 5,000,000 £/year, determine if this proposed investment is justified.

[6]